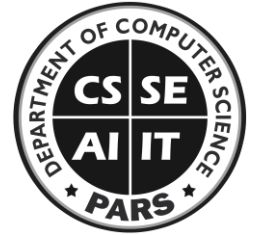




UNIVERSITY OF AGRICULTURE

Department of Computer Science



Lab 2: Programming Fundamentals

(CS-308)

Basic Information			
Lab #:	2	Teacher:	Arsal Mahmood
Objectives	1. Single selection in C++ 2. Double selection in C++ 3. Multiple selection in C++		
Registration#	2025-ag-9987	Student Name	Syed Rameez Haider Rizvi
Total Marks	10	Marks Obtained	
Tools	Dev C++ / Visual Studio		

Activity 1 : Single Selection (if statement)

1. Write a program that takes an integer from the user and checks whether the number is even or odd using an if statement.

```
#include <iostream>

using namespace std;

int main() {
    int number;

    cout << "Enter a number: ";
    cin >> number;

    if (number % 2 == 0) {
        cout << "The number is Even.";
    }
}
```

```

        if (number % 2 == 1) {    cout
<< "The number is Odd.";
    }
}

```

```

Enter a number: 1
The number is Odd.
-----
Process exited after 2.613 seconds with return value 0
Press any key to continue . . .

```

- Write a program that takes a number as input and determines whether the number is positive, negative, or zero.

```

#include <iostream>

using namespace

std; int main() {    int
number;    cout << "Enter
a number:
";    cin >> number;    if
(number == 0) {    cout <<
"You entered 0.";
    }
    if (number > 0) {
        cout << "You entered a positive integer.";
    }
    if (number < 0) {
        cout << "You entered a negative integer.";
    }
}

```

```

Enter a number: 4
You entered a positive integer.
-----
Process exited after 1.652 seconds with return value 0
Press any key to continue . . .

```

- Write a program that inputs three numbers from the user and determines which number is the largest using conditional statements.

```

#include <iostream> using
namespace std; int main()

```

```

{
    int a,b,c;

    cout<<"Enter 1st
Number:";

    cin>>a;

    cout<<"Enter 2nd Number:";

    cin>>b;

    cout<<"Enter 3rd Number:";    cin>>c;

    if(a>b && a>c) {        cout<<"1st
Number is Largest"<<endl;

        }

        if(b>c && b>a) {
            cout<<"2nd Number is Largest"<<endl;

        }

    if(c>b && c>a) {    cout<<"3rd Number is Largest"<<endl;

        }

    }

```

```

Enter 1st Number:5
Enter 2nd Number:6
Enter 3rd Number:7
3rd Number is Largest

-----
Process exited after 4.241 seconds with return value 0
Press any key to continue . . .

```

4. Write a program that takes a character from the user and checks whether it is a vowel or a consonant using logical operators.

```

#include <iostream> using
namespace std; int main()
{

```

```

char ch;  cout << "Enter a character: ";  cin >> ch;  if (!((ch
>= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))    cout <<
"Invalid input! Not an alphabet." << endl;  if ((ch == 'a' || ch
== 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||      ch == 'A' || ch
== 'E' || ch == 'I' || ch == 'O' || ch == 'U'))    cout << "It is a
Vowel." << endl;  if (((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch
<= 'Z')) &&      !(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o'
|| ch == 'u' ||      ch == 'A' || ch == 'E' || ch == 'I' || ch ==
'O' || ch == 'U'))    cout << "It is a Consonant." << endl;
}

```

```

Enter a character: a
It is a Vowel.

-----
Process exited after 1.722 seconds with return value 0
Press any key to continue . . .

```

5. Write a program that inputs a number and checks whether it is divisible by both 3 and 5 using logical operators. #include <iostream> using namespace std; int main()

```

{
int num;  cout << "Enter a number: ";  cin >> num;  if (num
% 3 == 0 && num % 5 == 0)    cout << "Number is divisible by
both 3 and 5." << endl;  if (!(num % 3 == 0 && num % 5 == 0))
cout << "Number is NOT divisible by both 3 and 5." << endl;
}

```

```

Enter a number: 15
Number is divisible by both 3 and 5.

-----
Process exited after 2.841 seconds with return value 0
Press any key to continue . . .

```

6. Write a program that inputs marks of 5 subjects of a student and calculate percentage of that student. Assigns a grade based on the following conditions:

- $\text{percentage} \geq 80 \rightarrow \text{Grade A}$
- $\text{percentage} \geq 70 \rightarrow \text{Grade B}$
- $\text{percentage} \geq 60 \rightarrow \text{Grade C}$
- $\text{percentage} \geq 50 \rightarrow \text{Grade D}$
- $\text{percentage} < 50 \rightarrow \text{Fail}$

```
#include <iostream>

using namespace std;

int main() {

    float m1, m2, m3, m4, m5, total, percentage;

    cout << "Enter marks of 5 subjects (0-100): ";

    cin >> m1 >> m2 >> m3 >> m4 >> m5;

    if ((m1 >= 0 && m1 <= 100) &&

(m2 >= 0 && m2 <= 100) &&

(m3 >= 0 && m3 <= 100) &&
(m4 >= 0 && m4 <= 100) && (m5

>= 0 && m5 <= 100)) {

    total = m1 + m2 + m3 + m4 + m5;

    percentage = total / 5;

    cout << "Percentage = " << percentage << "%" << endl;

    if (percentage >= 80)
        cout << "Grade A" << endl;

    if (percentage >= 70 && percentage < 80)

        cout << "Grade B" << endl;

    if (percentage >= 60 && percentage < 70)
```

```

        cout << "Grade C" << endl;

    if (percentage >= 50 && percentage < 60)

        cout << "Grade D" << endl;

    if (percentage < 50)

        cout << "Fail" << endl;

}

if (!((m1 >= 0 && m1 <= 100) &&

(m2 >= 0 && m2 <= 100) &&

(m3 >= 0 && m3 <= 100) &&

(m4 >= 0 && m4 <= 100) && (m5

>= 0 && m5 <= 100))) {

    cout << "Invalid input. Marks must be between 0 and 100." << endl;

}

}

```

```

Enter marks of 5 subjects (0-100): 80
90
80
70
89
Percentage = 81.8%
Grade A

-----
Process exited after 14.31 seconds with return value 0
Press any key to continue . . .

```

P-T-O

7. Write a program that inputs two numbers and displays which number is greater or displays "Both numbers are equal" if they are the same.

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {

    int a,b;

    cout<<"Enter 2 numbers:";

    cin>>a>>b;

    if(a>b)

        cout<<a<<" is greater";

    if(a<b)

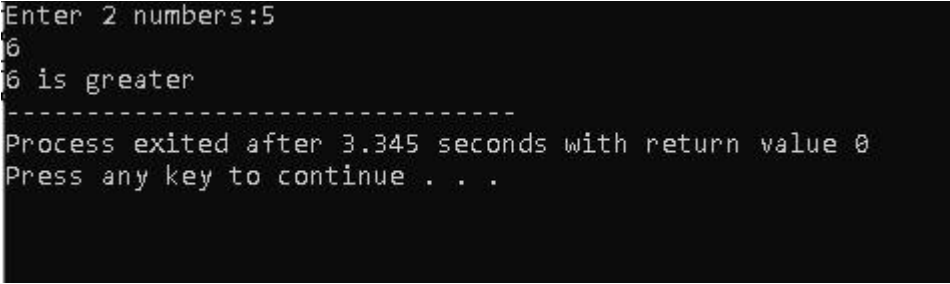
        cout<<b <<" is greater";

    if(a==b)

        cout<<"Both numbers are equal";

}

```



```

Enter 2 numbers:5
6
6 is greater
-----
Process exited after 3.345 seconds with return value 0
Press any key to continue . . .

```

Activity 2 : Double Selection (if-else statement, ternary operator)

- Write a program that inputs the age of a person and determines whether the person is eligible to vote ($\text{age} \geq 18$) or not eligible by using ternary operator.

```

#include    <iostream>

using namespace std; int

main() {    int    age;

    cout<<"Enter your age:";

    cin>>age;        string

    eligibility = (age>=18) ?

```

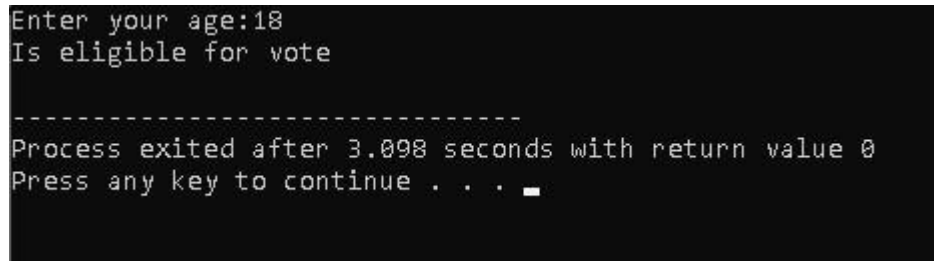
```

    "Is eligible" : "Not
eligible";

cout<<eligibility<<endl;

}

```



```

Enter your age:18
Is eligible for vote

-----
Process exited after 3.098 seconds with return value 0
Press any key to continue . . .

```

9. Write a program that takes a student's marks as input and displays "Pass" if marks are greater than or equal to 50, otherwise display "Fail" (use ternary operator).

```

#include <iostream>

using namespace std;

int main(){

    int marks;

    cout<<"Enter marks(0-100):";

    cin>>marks;

    if (marks>0 && marks<100){

        string result = (marks>=50) ? "Pass" : "Fail";

        cout<<result<<endl;

    }
    if (!(marks>0 && marks<100))

        cout<<"You entered Invalid numbers";

    }

```



```
Enter marks(0-100):89
Pass
-----
Process exited after 2.545 seconds with return value 0
Press any key to continue . . .
```

Activity 3 : Multiple Selection (else ifs statement)

10. Write a program that inputs a person's age and displays whether the person is a child (age < 13), teenager (13–19), or adult (20 and above).

```
#include <iostream>
using namespace std; int
main(){ int age;
cout<<"Enter your age:";
cin>>age; if(age<13)
cout<<"You are a child"; else
if(age>13 && age<=19)
cout<<"You are a Teenager";
else
cout<<"You are a adult";
}
```

```
Enter your age:19
Your are a Teenager
-----
Process exited after 6.68 seconds with return value 0
Press any key to continue . . .
```

11. Write a program that inputs a number and checks whether the number is divisible by both 3 and 5, only by 3, only by 5, or none of them.

```
#include <iostream>
using namespace std; int
main(){ int num;
cout<<"Enter a num:";
cin>>num;
if(num%5==0 && num%3==0)
cout<<"divisible by both 3 and 5";
else if(num%3==0)
cout<<"divisible by 3"; else
if(num%5==0) cout<<"divisible by
```

```
5"; else
cout<<"None of them";
}
```

```
Enter a num:10
divisible by 5
-----
Process exited after 2.202 seconds with return value 0
Press any key to continue . . .
```

12. Write a program that asks the user to enter three numbers and determines whether all numbers are equal, two numbers are equal, or all numbers are different.

```
#include <iostream>
using namespace
std; int main(){ int
a,b,c;
cout<<"Enter 3 numbers:";
cin>>a>>b>>c; if (a==b && a==c)
cout<<"all numbers are equal";
else if (a==b || a==c)
cout<<"two numbers are
equal"; else
cout<<"all numbers are different";
}
```

```
Enter 3 numbers:5
8
4
all numbers are different
-----
Process exited after 4.192 seconds with return value 0
Press any key to continue . . .
```

13. Write a program that inputs the age of a person and determines whether the person is not eligible for voting (<18), eligible (18–60), or a senior citizen (>60).

```
#include <iostream> using
namespace std; int main(){ int
age; cout<<"Enter age:";
cin>>age; if (age<18)
cout<<"Not eligible"; else
if(age>=18 && age<=60)
cout<<"Eligible"; else
cout<<"Senior citizen";
```

```
}
```

```
Enter age:19
Eligible for vote
-----
Process exited after 2.877 seconds with return value 0
Press any key to continue . . .
```

14. Write a program that asks the user to enter the marks of a student and displays Grade A (≥ 85), Grade B (70–84), Grade C (50–69), or Fail (< 50). Use logical operators.

```
#include <iostream>
using namespace std;
int main(){ float marks;
    cout << "Enter marks of the student: ";
    cin >> marks;  if (marks >= 85)
    cout << "Grade A" << endl;  else if
    (marks >= 70 && marks < 85)    cout
    << "Grade B" << endl;  else if (marks >=
    50 && marks < 70)    cout << "Grade
    C" << endl;  else    cout << "Fail" <<
    endl;
}
```

```
Enter marks of the student: 78
Grade B
-----
Process exited after 3.939 seconds with return value 0
Press any key to continue . . .
```

15. Write a program that asks the user to enter the lengths of three sides of a triangle. The program should then determine the type of triangle based on the equality of its sides.

- If all three sides are equal, the triangle is called an Equilateral Triangle.
- If exactly two sides are equal, the triangle is called an Isosceles Triangle.
- If all three sides are different, the triangle is called a Scalene Triangle.

The program should display the type of triangle according to the entered side lengths.

```
#include <iostream>
using namespace
std; int main(){ float
a,b,c;
    cout<<"Enter the
    length:"; cin>>a>>b>>c; if
    (a==b && a==c)
```

```

cout<<"Its Equilateral Triangle";
else if (a==b || a==c) cout<<"Its
Isosceles Triangle"; else
cout<<"Its Scalene Triangle";
}

```

```

Enter the length:34
34
23
Its Isosceles Triangle
-----
Process exited after 4.727 seconds with return value 0
Press any key to continue . . .

```

16. Write a program that takes a single character as input from the user. The program should determine the type of the character using conditional statements.

- If the character is between '0' and '9', it should be classified as a Digit.
- If the character is between 'A'-'Z' or 'a'-'z', it should be classified as an Alphabet.
- Any other character should be classified as a Special Character.

The program should display the appropriate category of the character.

```

#include <iostream>
using namespace
std; int main(){ char
a;
cout<<"Enter a
Character:"; cin>>a; if
(a>='0' && a<='9')
cout<<"Its a digit";
else if ((a>='A' && a<='Z') || (a>='a' &&
a<='z')) cout<<"Its an alphabet"; else
cout<<"special character";
}

```

```

Enter a Character:a
Its an alphabet
-----
Process exited after 1.882 seconds with return value 0
Press any key to continue . . .

```

17. Write a program that asks the user to enter the basic salary of an employee. The program should calculate the net salary by adding an allowance according to the following conditions:

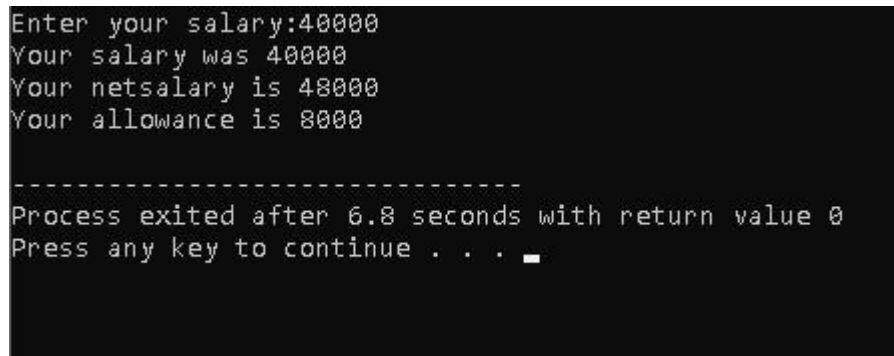
- If the basic salary is less than 50,000, the employee receives an allowance of 20% of the salary.
- If the basic salary is 50,000 or greater, the employee receives an allowance of 10% of the salary.

The program should then calculate the net salary using the formula:

Net Salary = Basic Salary + Allowance

Finally, the program should display the allowance amount and the net salary.

```
#include <iostream>
using namespace std; int
main(){
int salary,netsalary,allowance; cout<<"Enter
your salary:"; cin>>salary; if (salary<50000)
allowance=salary*20/100; if (salary>=50000)
allowance=salary*10/100;
netsalary=salary+allowance; cout<<"Your
salary was "<<salary<<endl; cout<<"Your
netsalary is "<<netsalary<<endl; cout<<"Your
allowance is "<<allowance<<endl;
}
```



```
Enter your salary:40000
Your salary was 40000
Your netsalary is 48000
Your allowance is 8000
-----
Process exited after 6.8 seconds with return value 0
Press any key to continue . . .
```

18. Develop a calculator which will calculate Sum, minus, multiplication and division of two user given numbers. It will calculate subtraction only when first number is greater than second number otherwise it should display *cannot subtract*. Same rule is applied for division. If the second number is 0, do not divide both numbers, instead display *cannot divide a number by 0*.

```
#include <iostream> using
namespace std; int main(){ int
num1,num2,n; cout<<"Enter
2 numbers:";
cin>>num1>>num2;
```

```

cout<<"Enter 1 for Sum:"<<endl<<"Enter 2 for Subtraction:"<<endl<<"Enter 3 for
Mulltiplication:"<<endl<<"Enter 4 for Division:"<<endl; cin>>n;
if(n==1)
cout<<"Sum is "<<num1+num2<<endl; else if(n==2)
{
if (num1>=num2)
cout<<"Subtract is "<<num1-num2<<endl; else
cout<<"cannot subtract";
}
else if (n==3)
cout<<"Multiple is "<<num1*num2<<endl; else if
(n==4)
{
if (num2==0)
cout<<"cannot divide a number by 0"<<endl; else
cout<<"division is "<<num1/num2<<endl;
}
else
cout<<"Invalid Number!"<<endl;
}

```

```

Enter 2 numbers:7
4
Enter 1 for Sum:
Enter 2 for Subtraction:
Enter 3 for Mulltiplication:
Enter 4 for Division:
2
Subtract is 3

-----
Process exited after 5.033 seconds with return value 0
Press any key to continue . . .

```

19. Ask the user to enter two numbers and then ask to enter the operator (+, -, /, *).

Calculate and display the result (use multiple selection statements).

```

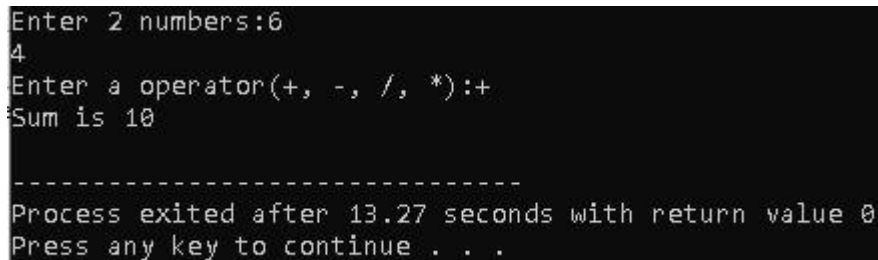
#include <iostream> using
namespace std; int
main(){ int num1,num2;
cout<<"Enter 2
numbers:";
cin>>num1>>num2;
char n;

```

```

cout<<"Enter a operator(+, -, /,
*):"; cin>>n; if(n=='+')
cout<<"Sum is "<<num1+num2<<endl;
else if(n=='-')
{
if (num1>=num2)
cout<<"Subtract is "<<num1-num2<<endl;
else
cout<<"cannot subtract";
}
else if (n=='*')
cout<<"Multiple is "<<num1*num2<<endl;
else if (n=='/')
{
if (num2==0)
cout<<"cannot divide a number by 0"<<endl;
else
cout<<"division is "<<num1/num2<<endl;
}
else
cout<<"Invalid Input!"<<endl;
}

```



```

Enter 2 numbers:6
4
Enter a operator(+, -, /, *):+
Sum is 10

-----
Process exited after 13.27 seconds with return value 0
Press any key to continue . . .

```

20. A bank gives loan in two situations

- If customer is male and his age is more than 25 and his salary is more than 25000 • If customer is a female and her age is more than 30 and her income is 20000

Program: Ask user to enter age, gender(M/F) and income. Check whether he\she is eligible to get loan or not.

```

#include <iostream> using namespace std; int main(){

int age,income; cout<<"Enter Your Age:"; cin>>age;

```

```

cout<<"Enter Your income:"; cin>>income; char g;

cout<<"Enter your Gender:"; cin>>g; if ((g=='M' ||

g=='m') && (age>25) && (income>25000)) cout<<"You

are eligible for Loan"; else if ((g=='F' || g=='f') &&

(age>30) && (income>20000)) cout<<"You are eligible

for Loan"; else cout<<"You are not eligible for Loan";

}

```

```

Enter Your Age:26
Enter Your income:26000
Enter your Gender:M
You are eligible for Loan
-----
Process exited after 18.64 seconds with return value 0
Press any key to continue . . .

```

(END OF LAB)